📘 Python Data Structures Notes (List, Tuple, Set, Dictionary) – With Output

🔹 1️⃣ LIST in Python

✅ Definition (One Line)

List ek ordered, mutable (changeable) collection hota hai jo multiple values store karta hai.
A list is an ordered, mutable collection used to store multiple values.

my_list = [10, 20, 30, "Hello", True]

print(my_list)

🖨️ **Output:**

[10, 20, 30, 'Hello', True]

⭐ Properties

- Ordered hoti hai
- Mutable (change ho sakti hai)
- Duplicate values allow hoti hain
- Indexing supported

📌 List Methods (With Output)

Method	One Line Definition	Example	Output
append()	List ke end me value add karta hai	a=[1,2]; a.append(3); print(a)	[1, 2, 3]
extend()	Ek list me dusri list add karta hai	a=[1]; a.extend([2,3]); print(a)	[1, 2, 3]
insert()	Specific index par value add karta hai	a=[1,3]; a.insert(1,2); print(a)	[1, 2, 3]
remove()	Given value ko remove karta hai	a=[1,2,3]; a.remove(2); print(a)	[1, 3]
pop()	Last element delete karta hai	a=[1,2]; a.pop(); print(a)	[1]
clear()	List ko empty karta hai	a=[1,2]; a.clear(); print(a)	[]

Method	One Line Definition	Example	Output
index()	Value ka index batata hai	a=[10,20]; print(a.index(20))	1
count()	Value kitni baar hai count karta hai	a=[1,1,2]; print(a.count(1))	2
sort()	List ko sort karta hai	a=[3,1,2]; a.sort(); print(a)	[1, 2, 3]
reverse()	List ko ulta karta hai	a=[1,2]; a.reverse(); print(a)	[2, 1]
copy()	List ka copy banata hai	b=[1,2]; c=b.copy(); print(c)	[1, 2]

🔹 2️⃣ TUPLE in Python

✅ Definition (One Line)

Tuple ek ordered, immutable (unchangeable) collection hota hai.
A tuple is an ordered, immutable collection of elements.

my_tuple = (10, 20, 30)

print(my_tuple)

🖨️ Output:

(10, 20, 30)

📌 Tuple Methods (With Output)

Method	One Line Definition	Example	Output
count()	Value kitni baar hai count karta hai	t=(1,1,2); print(t.count(1))	2
index()	Value ka index deta hai	t=(10,20); print(t.index(20))	1

🔹 3️⃣ SET in Python

✅ Definition (One Line)

Set ek unordered collection hota hai jisme unique values hoti hain.
A set is an unordered collection of unique elements.

my_set = {10, 20, 30}

print(my_set)

🖨️ **Output:**

{10, 20, 30}

(Order change ho sakta hai)

📌 **Set Methods (With Output)**

Method	One Line Definition	Example	Output
add()	Set me element add karta hai	s={1,2}; s.add(3); print(s)	{1, 2, 3}
update()	Multiple values add karta hai	s={1}; s.update([2,3]); print(s)	{1, 2, 3}
remove()	Element remove karta hai	s={1,2}; s.remove(2); print(s)	{1}
discard()	Element remove karta hai (error nahi deta)	s={1,2}; s.discard(5); print(s)	{1, 2}
pop()	Random element delete karta hai	s={1,2}; s.pop(); print(s)	{1} or {2}
clear()	Set empty karta hai	s={1,2}; s.clear(); print(s)	set()
union()	Do set ko join karta hai	a={1,2}; b={2,3}; print(a.union(b))	{1, 2, 3}
intersection()	Common elements deta hai	print(a.intersection(b))	{2}
difference()	Difference deta hai	print(a.difference(b))	{1}

🔹 4️⃣ **DICTIONARY in Python**

✅ **Definition (One Line)**

Dictionary ek key-value pair ka collection hota hai.
A dictionary stores data in key-value pairs.

my_dict = {"name": "Rahul", "age": 20}

print(my_dict)

🖨️ **Output:**

{'name': 'Rahul', 'age': 20}

📌 **Dictionary Methods (With Output)**

Method	One Line Definition	Example	Output
get()	Key ki value deta hai	d={"name":"Aman"}; print(d.get("name"))	Aman

Method	One Line Definition	Example	Output
keys()	All keys deta hai	print(d.keys())	dict_keys(['name'])
values()	All values deta hai	print(d.values())	dict_values(['Aman'])
items()	Key-value pair deta hai	print(d.items())	dict_items([('name','Aman')])
update()	Dictionary update karta hai	d.update({"age":22}); print(d)	{'name':'Aman','age':22}
pop()	Key ke through value delete karta hai	d.pop("age"); print(d)	{'name':'Aman'}
popitem()	Last item delete karta hai	d.popitem(); print(d)	{}
clear()	Dictionary empty karta hai	d.clear(); print(d)	{}
copy()	Dictionary ka copy banata hai	c=d.copy(); print(c)	{}

🔥 One Program – All in One (With Output)

lst = [1, 2, 3]

lst.append(4)

print(lst)

*

tup = (10, 20, 30)

print(tup.count(10))

st = {1, 2, 3}

st.add(4)

print(st)

*

d = {"name": "Aman", "age": 21}

d["city"] = "Delhi"

print(d)

🖨️ Output:

[1, 2, 3, 4]

1

{1, 2, 3, 4}

{'name': 'Aman', 'age': 21, 'city': 'Delhi'}
